



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22MA201

ENGINEERING MATHEMATICS II

UNIT 4 & LP3 –GREEN’S THEOREM

Green's Theorem

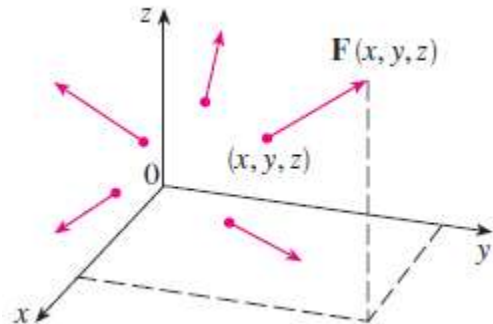
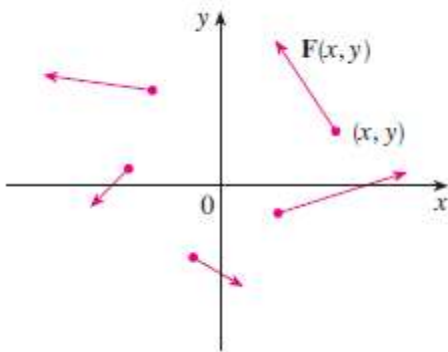
Vector Field:

A vector field in the plane is a function $F(x, y)$ mapping points in $\mathbb{R}^2$ into the set of two-dimensional vectors V_2 . We write

$$F(x, y) = f_1(x, y)\vec{i} + f_2(x, y)\vec{j}$$

Similarly, the vector field in the three dimensional plane $\mathbb{R}^3$ is of the form

$$F(x, y, z) = f_1(x, y, z)\vec{i} + f_2(x, y, z)\vec{j} + f_3(x, y, z)\vec{k}$$

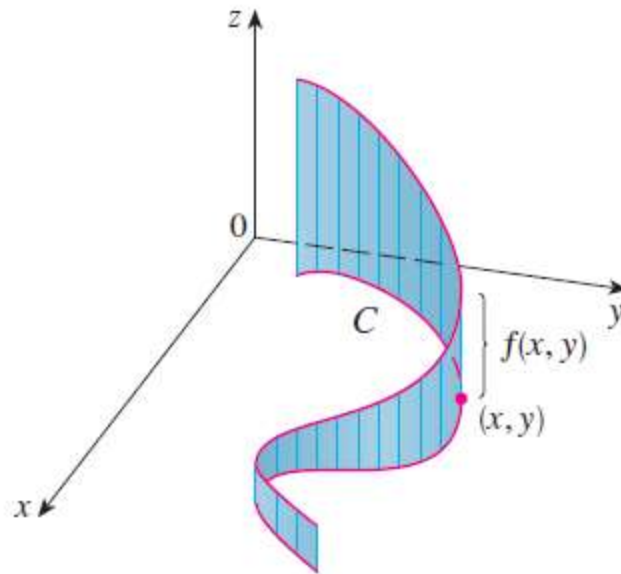


Line Integral:

Let $F(x, y) = M(x, y)\vec{i} + N(x, y)\vec{j}$ be a vector field then the line integral of F over the curve C is defined by

$$\int_C F \cdot \vec{dr} = \int_C M(x, y) dx + N(x, y) dy$$

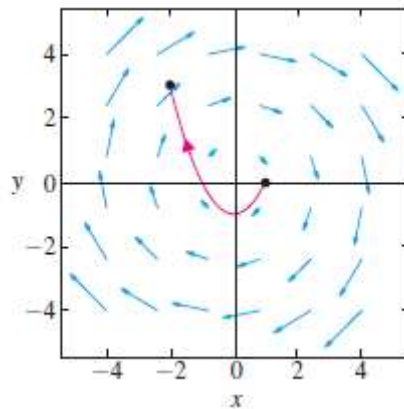
where $\vec{dr} = dx\vec{i} + dy\vec{j}$.



Problems:

1. If $F(x, y) = y\vec{i} - x\vec{j}$ and C is the parabola $y = x^2 - 1$ from $(1, 0)$ to $(-2, 3)$ then evaluate the line integral $\int_C F \cdot d\vec{r}$.

Solution:



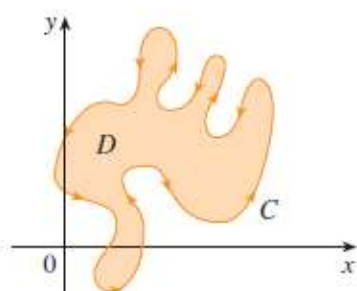
Given $y = x^2 - 1 \Rightarrow dy = 2x dx$

Also x varies from 1 to -2

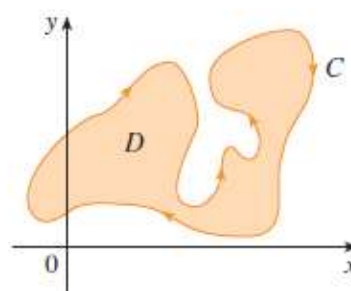
$$\begin{aligned}
\int_C \mathbf{F} \cdot d\vec{r} &= \int_C M(x, y) dx + N(x, y) dy \\
&= \int_C y dx - x dy = \int_1^{-2} (x^2 - 1) dx - x(2x dx) \\
&= \int_1^{-2} (-x^2 - 1) dx = 6
\end{aligned}$$

Green's Theorem:

In stating Green's Theorem we use the convention that the positive orientation of a simple closed curve C refers to a single counterclockwise traversal of C .



(a) Positive orientation



(b) Negative orientation

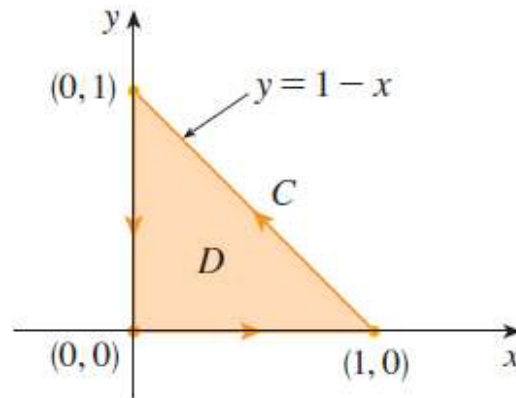
Green's Theorem:

Let C be a piecewise-smooth, simple closed curve in the plane with positive orientation and let R be the region enclosed by C , together with C . Suppose that $M(x, y)$ and $N(x, y)$ are continuous and have continuous first partial derivatives in some open region D , with $R \subset D$. Then,

$$\oint_C M(x, y) dx + N(x, y) dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA.$$

Problems:

1. Evaluate $\int_C x^4 dx + xy dy$ where C is the triangular curve consisting of the line segments from $(0,0)$ to $(1,0)$, from $(1,0)$ to $(0,1)$, and from $(0,1)$ to $(0,0)$.

Solution:

Here $M(x,y) = x^4$, $N(x,y) = xy$

X – limits : 0 to 1

Y – limits : 0 to $1-x$

Therefore, by Green's Theorem we have

$$\begin{aligned}
 \int_C x^4 dx + xy dy &= \iint_D \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA = \int_0^1 \int_0^{1-x} (y - 0) dy dx \\
 &= \int_0^1 \left[\frac{1}{2} y^2 \right]_{y=0}^{y=1-x} dx = \frac{1}{2} \int_0^1 (1-x)^2 dx \\
 &= -\frac{1}{6} (1-x)^3 \Big|_0^1 = \frac{1}{6}
 \end{aligned}$$

Verification:

In this problem $\int_C = \int_{C_1} + \int_{C_2} + \int_{C_3}$

Along the curve C_1 :

$$y = 0, \quad dy = 0$$

x varies from 0 to 1

$$\int_{C_1} x^4 dx + xy \, dy = \int_0^1 x^4 dx = \left[\frac{x^5}{5} \right]_0^1 = \frac{1}{5}$$

Along the curve C_2 :

$$y = 1 - x, \quad dy = -dx$$

x varies from 1 to 0

$$\int_{C_2} x^4 dx + xy \, dy = \int_1^0 x^4 dx + x(1-x)(-dx)$$

$$= \int_1^0 (x^2 - x + x^4) dx = -\frac{1}{30}$$

Along the curve C_3 :

$$x = 0, \quad dx = 0$$

y varies from 1 to 0

$$\int_{C_3} x^4 dx + xy \, dy = \int_1^0 0 = 0$$

Hence,
$$\int_C x^4 dx + xy \, dy = \frac{1}{5} - \frac{1}{30} + 0 = \frac{1}{6}$$

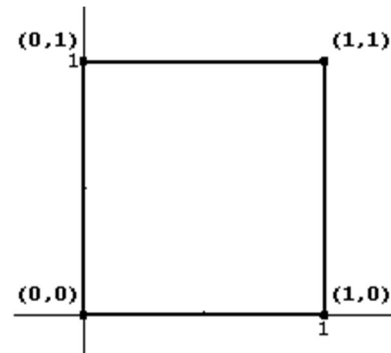
2. Evaluate the line integral $\int_C xy \, dy - y^2 \, dx$ where C is the square cut from the first quadrant by the lines $x = 1$ and $y = 1$.

Solution:

Here $M(x, y) = -y^2$, $N(x, y) = xy$

X – limits : 0 to 1

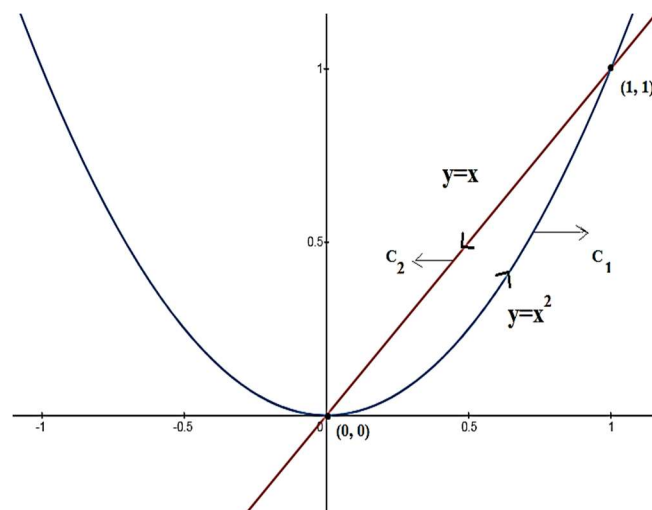
Y – limits : 0 to 1



$$\oint_C -y^2 \, dx + xy \, dy = \iint_R (y - (-2y)) \, dx \, dy = \int_0^1 \int_0^1 3y \, dx \, dy$$

$$= \int_0^1 \left[3xy \right]_{x=0}^{x=1} dy = \int_0^1 3y \, dy = \left[\frac{3}{2}y^2 \right]_0^1 = \frac{3}{2}.$$

3. Use Green's Theorem to evaluate $\int_C (xy + y^2) \, dx + x^2 \, dy$ where C is the closed curve of the region bounded by the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$, followed by the line $y = x$ from $(1, 1)$ to $(0, 0)$ as shown in the Fig.



Solution:

Here $M(x, y) = xy + y^2$, $N(x, y) = x^2$

X – limits : 0 to 1

Y – limits : x^2 to x

Therefore, by Green's Theorem we have

$$\begin{aligned}\int_C (xy + y^2) dx + x^2 dy &= \int_0^1 \int_{x^2}^x \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx \\ &= \int_0^1 \int_{x^2}^x (2x - x - 2y) dy dx = \int_0^1 \int_{x^2}^x (x - 2y) dy dx \\ &= \int_0^1 (x^3 - x^4) dx = \frac{-1}{20}\end{aligned}$$

Verification:

In this problem $\int_C = \int_{C_1} + \int_{C_2}$

Along the curve C_1 :

$$y = x^2, \quad dy = 2x dx$$

x varies from 0 to 1

$$\int_{C_1} (xy + y^2) dx + x^2 dy = \int_0^1 (x^4 + 3x^3) dx = \left[\frac{x^5}{5} + \frac{3x^4}{4} \right]_0^1 = \frac{19}{20}$$

Along the curve C_2 :

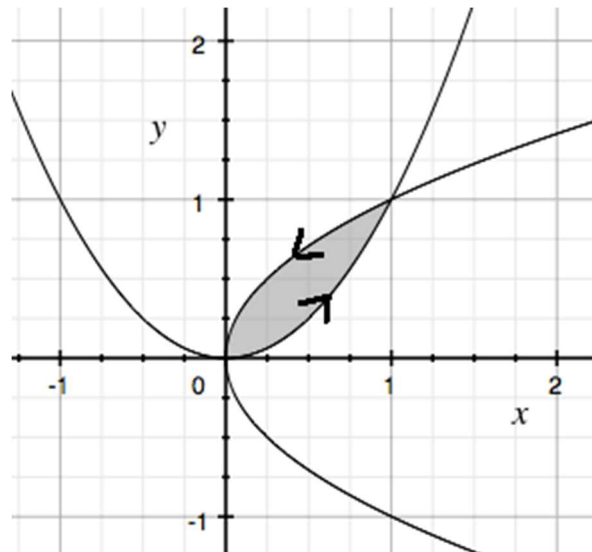
$$y = x, \quad dy = dx$$

x varies from 1 to 0

$$\int_{C_2} (xy + y^2) dx + x^2 dy = \int_1^0 3x^2 dx = \left[x^3 \right]_1^0 = -1$$

$$\text{Hence,} \quad \int_C (xy + y^2) dx + x^2 dy = \frac{19}{20} - 1 = \frac{-1}{20}$$

4. Use Green's Theorem to evaluate $\int_C (x^2 - y^2) dx + 2xy dy$ where C is where C is the closed curve of the region bounded $y = x^2$ and $x = y^2$ as shown in the Fig.



Solution:

$$\text{Here } M(x, y) = x^2 - y^2, \quad N(x, y) = 2xy$$

X – limits : 0 to 1

Y – limits : x^2 to $\sqrt{x}$

Therefore, by Green's Theorem we have

$$\int_C (x^2 - y^2) dx + 2xy dy = \int_0^1 \int_{x^2}^{\sqrt{x}} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

$$= \int_0^1 \int_{x^2}^{\sqrt{x}} (4y) dy dx$$

$$= \int_0^1 (2x^4 - 2x) dx = \frac{3}{5}$$

Verification:

$$\text{In this problem} \quad \int_C = \int_{C_1} + \int_{C_2}$$

Along the curve C_1 :

$$y = x^2, \quad dy = 2x dx$$

x varies from 0 to 1

$$\int_{C_1} (x^2 - y^2) dx + 2xy dy = \int_0^1 (x^2 + 3x^4) dx = \left[\frac{x^3}{3} + \frac{3x^5}{5} \right]_0^1 = \frac{1}{3} + \frac{3}{5}$$

Along the curve C_2 :

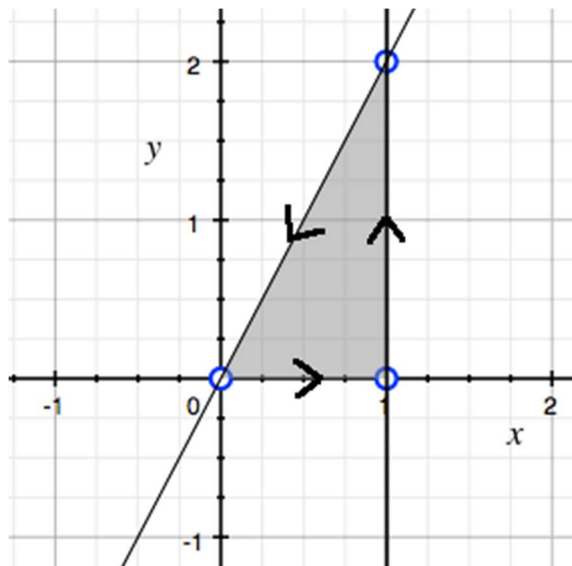
$$x = y^2, \quad dx = 2y dy$$

y varies from 1 to 0

$$\int_{C_2} (x^2 - y^2) dx + 2xy dy = \int_1^0 2y^5 dy = \left[\frac{y^6}{3} \right]_1^0 = -\frac{1}{3}$$

Hence,
$$\int_C (x^2 - y^2) dx + 2xy dy = \frac{1}{3} + \frac{3}{5} - \frac{1}{3} = \frac{3}{5}$$

5. Use Green's Theorem to evaluate $\int_C xy dx + x^2 y^3 dy$ where C is the positively oriented triangle whose vertices are (0,0), (1,0), and (1,2).



Solution:

$$\text{Here } M(x, y) = xy, \quad N(x, y) = x^2 y^3$$

X – limits : 0 to 1

Y – limits : 0 to 2x

Therefore, by Green's Theorem we have

$$\begin{aligned}\oint_C xy \, dx + x^2 y^3 \, dy &= \iint_D (2xy^3 - x) \, dA \\ \oint_C xy \, dx + x^2 y^3 \, dy &= \int_0^1 \int_0^{2x} (2xy^3 - x) \, dy \, dx \\ \oint_C xy \, dx + x^2 y^3 \, dy &= \int_0^1 \left[\frac{xy^4}{2} - xy \right]_{y=0}^{y=2x} dx \\ \oint_C xy \, dx + x^2 y^3 \, dy &= \int_0^1 (8x^5 - 2x^2) \, dx \\ \oint_C xy \, dx + x^2 y^3 \, dy &= \left[\frac{8x^6}{6} - \frac{2x^3}{3} \right]_{x=0}^{x=1} \\ \oint_C xy \, dx + x^2 y^3 \, dy &= \frac{8}{6} - \frac{2}{3} \\ \oint_C xy \, dx + x^2 y^3 \, dy &= \frac{2}{3}\end{aligned}$$

Discussion Questions:

1. What are real-world applications of Green's Theorem?
2. Interpret the geometric interpretation of Green's Theorem?
3. Mention the limitations of Green's Theorem?
4. How does Green's Theorem extend to three dimensions?